

Maurice D. Hanisch

Pasadena, CA

Google Scholar: MD Hanisch

Email: mhanisch@caltech.edu

LinkedIn: linkedin.com/in/MauriceHanisch

Mobile: (626) 394-7713

Education

California Institute of Technology, *Ph.D. Math & CS*

Current GPA: 3.94

09/2025 – Present

- Researching machine learning for quantum chemistry and materials science with Prof. Anima Anandkumar.
- Fully funded through the Kortschak Fellowship, awarded based on academic excellence and research potential.

ETH Zurich, *M.Sc. Physics*

GPA: 5.77/6.00 (Top 12%)

09/2022 – 05/2024

- Expanded analytical skills with a focus on quantum theory and information technologies, with 30% excess credits in mathematics, ML and probabilistic AI involving 8 team projects, all graded 6.00/6.00 (🟢 projects).

LMU Munich, *B.Sc. Physics*

GPA: 1.28/1.00 (Top 10%)

10/2018 – 09/2022

- Acquired extensive grounding in key physics fields, providing a strong foundation in the principles driving modern technologies, and fostering advanced analytical and problem-solving skills.

Work Experience

Bloomberg CTO Office, *Security Research Intern, Julien Vanegue's Team*

06/2025 – 09/2025

- Stress-tested the CHERI CPU architecture, a \$300 million industry-academic security effort, by attempting to break its memory-safety guarantees to determine its suitability for Bloomberg's infrastructure.
- Discovered multiple attack vectors that compromise both spatial and temporal memory safety guarantees.
- Contributed mitigations to the open-source CHERI software via multiple PRs (🟢 jemalloc, 🟢 dlmalloc, 🟢 c-guide).
- Co-authored a paper accepted at ACM CCS 2026.

California Institute of Technology, *Machine Learning Research Intern, Anima Anandkumar's Group*

09/2024 – 05/2025

- Led a large-scale benchmark, retraining seven state-of-the-art molecular models for open-shell and charged systems, validating our proposed method, *OrbitAll*, in our published work.
- Developed a differentiable quantum feature generation pipeline for graph neural network (GNN) models, enabling end-to-end differentiability of physics-informed learning of quantum chemical properties.

Research Experience

IBM, *Master's Thesis, Stefan Woerner's Group* | 📄 Thesis (PDF), 📁 Project repository

Grade: 6.00/6.00

09/2023 – 05/2024

- Developed a Python/C++ decoding pipeline that scaled to billions of measurement records per run, enabling one of IBM's then-largest error-correction experiments.
- Designed and deployed the full experimental pipeline on IBM Q hardware, showing that analog-information decoding can improve logical error rates by up to 30x in large-scale repetition-code experiments.

ETH Zurich, *Summer Project, Jonathan Home's Group* | 📄 Thesis (PDF)

06/2023 – 08/2023

- Investigated the motional interaction of ions trapped in an electromagnetic field for use in quantum computing.
- Designed and simulated trapping potentials using Python to enable a beamsplitter interaction.
- Experimentally validated the designed potentials on GKP-encoded qubits within a trapped-ion setup.

MPI for Quantum Optics, *Bachelor's Thesis, Ignacio Cirac's Group* | 📄 Thesis (PDF)

Grade: 1.30/1.00

04/2022 – 09/2022

- Investigated efficient algorithm design and resource optimization for Gaussian quantum circuits.
- Applied complexity theory and mathematical techniques to prove that bosonic Gaussian circuits can be implemented on quantum computers with exponentially fewer qubits compared to a direct encoding.

Selected Publications

Learning the Kohn-Sham map with neural operators for quasi-linear scaling density functional theory

- D. Khan^{*}, M.D. Hanisch^{*}, N. Argatoff, E. Xie, S. Sharma, A. Anandkumar - arXiv:2608.23895 (2026)

Soft information decoding with superconducting qubits

- M.D. Hanisch, B. Hetényi, J.R. Wootton - APS Open Sci. 1, 000019 (2026)

MGB: The Material Generation Benchmark

- L. Yan, B.S. Kang, M.D. Hanisch, J. Ma, A. Anandkumar - NeurIPS AI4Mat Workshop (2025)

OrbitAll: A Unified Quantum Mechanical Representation Deep Learning Framework for All Molecular Systems

- B.S. Kang, V.C. Bhethanabotla, A. Tavakoli, M.D. Hanisch, ..., A. Anandkumar - arXiv:2507.03853 (2025)

Extracurricular Activities and Competitions

- Participated in 10+ technical competitions and workshops across cities like Paris, Munich, Zurich and New York, sponsored by companies ranging from consulting firms (Bain, McKinsey) to tech giants (Bloomberg, Infineon, NVIDIA, EDF).
- Won over \$3500 in hackathon prizes for projects using recurrent NNs, time series, LLMs, CUDA-Q, SageMaker and Hugging Face.

Hackathons: 1st/30 at ETH QHack 2024: 🟢 NVIDIA task | 1st/12 at LOQC 2023: 🟢 EDF task | 2nd/6 at ETH QHack 2023: 🟢 IQM task

Skills

Programming: Python, C, C++, Fortran, Matlab, SQL | PyTorch, TensorFlow, Scikit-learn | Git, Linux, Latex, Docker

Languages: German (native), French (native), English (C1: IELTS 8/9), Italian (A1)

Interests: Weightlifting, Volleyball, Calisthenics, Surfing, Chess